

Leaching kinetics of roasted zinc concentrate in sulfuric acid solutions

Estudo cinético da lixiviação de concentrado de zinco ustulado em soluções de ácido sulfúrico

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ABSTRACT

This work investigates the leaching kinetics of roasted zinc in aqueous solutions of sulphuric acid from an electrolytic zinc production plant. Parameters such as stirring speed, temperature, granulometry, and acid concentration were investigated aiming to model the experimental data according to the shrinking core and progressive conversion models. The kinetic study determined that leaching of the roasted concentrate follows the progressive conversion model and is controlled by the chemical reaction step, whose activation energy, enthalpy, and apparent entropy values are $13.45 \text{ kJ}\cdot\text{mol}^{-1}$, $11.07 \text{ kJ}\cdot\text{mol}^{-1}$ and $-256.20 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, respectively. The estimated value of the chemical reaction rate constant (k_s) is $3.00 \times 10^{-2} \text{ cm}\cdot\text{s}^{-1}$.

Keywords: Kinetic Study, Roasted Zinc Concentrate, Shrinking Core Model, Progressive Conversion Model, Leaching.

RESUMO

Este trabalho investiga a cinética de lixiviação do concentrado de zinco ustulado em soluções aquosas de ácido sulfúrico provenientes de uma planta de produção de zinco eletrolítico. Foram avaliados parâmetros como velocidade de agitação, temperatura, granulometria e concentração de ácido, com o objetivo de modelar os dados experimentais de acordo com os modelos do núcleo em diminuição e da conversão progressiva. O estudo cinético determinou que a lixiviação do concentrado ustulado segue o modelo de conversão progressiva e é controlada pela etapa de reação química, cujos valores de energia de ativação, entalpia e entropia aparente são, respectivamente, $13,45 \text{ kJ}\cdot\text{mol}^{-1}$, $11,07 \text{ kJ}\cdot\text{mol}^{-1}$ e $-256,20 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$. O valor estimado da constante de velocidade da reação química (k_s) é $3,00 \times 10^{-2} \text{ cm}\cdot\text{s}^{-1}$.

Palavras-chave: Estudo Cinético, Concentrado Ustulado de Zinco, Modelo do Núcleo em Diminuição, Modelo de Conversão Progressiva, Lixiviação.

RESUMEN

Este trabajo investiga la cinética de lixiviación del concentrado de zinc calcinado en soluciones acuosas de ácido sulfúrico provenientes de una planta de producción de zinc electrolítico. Se evaluaron parámetros como la velocidad de agitación, la temperatura, la

granulometría y la concentración de ácido, con el objetivo de modelar los datos experimentales de acuerdo con los modelos de núcleo en contracción y de conversión progresiva. El estudio cinético determinó que la lixiviación del concentrado calcinado sigue el modelo de conversión progresiva y está controlada por la etapa de reacción química, cuyos valores de energía de activación, entalpía y entropía aparente son, respectivamente, $13,45 \text{ kJ}\cdot\text{mol}^{-1}$, $11,07 \text{ kJ}\cdot\text{mol}^{-1}$ y $-256,20 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$. El valor estimado de la constante de velocidad de la reacción química (k_s) es $3,00 \times 10^{-2} \text{ cm}\cdot\text{s}^{-1}$.

Palabras clave: Estudio Cinético, Concentrado Calcinado de Zinc, Modelo de Núcleo en Contracción, Modelo de Conversión Progresiva, Lixiviación.

1 INTRODUCTION

The dissolution of ZnO in sulphuric acid solution has been investigated by several authors from both practical and theoretical viewpoints (Bortot Coelho *et al.*, 2020; Coelho *et al.*, 2018; Kashyap and Taylor, 2022; USGS, 2025). Jandová *et al.* (1999) investigated the leaching behaviour of several ZnO compounds obtained by calcination of zinc hydroxides produced by precipitation of ZnSO_4 solutions. They concluded that ZnO leaching in diluted H_2SO_4 solutions is controlled by diffusion processes. Terry (1983) established correlations involving chemical reaction rates of ZnO, Zn_2SiO_4 and ZnFe_2O_4 individual species and of the different mixture of these compounds in aqueous solutions of H_2SO_4 . Doyle *et al.* (1987) developed a mathematical model to describe the leaching kinetics of ZnO in diluted sulphuric acid solutions considering that the mass transport controls the dissolution rate.

Most studies that seek to understand the dissolution mechanisms use pure species or a mixture of synthetic compounds. However, the approach involving dissolution aspect of roasted zinc concentrates in sulphuric medium obtained in industrial scale is scarce and is relevant as 80% of the zinc presently produced in the world is produced by roasting-leaching-electrolysis using zinc sulphide concentrates and, in some cases, associated with the leaching of zinc silicates (Chen and Dutrizac, 2004; Harmer *et al.*, 2007; Souza *et al.*, 2007; Balarini *et al.*, 2008).

The literature presents several kinetic models that describe the reaction between a solid and a fluid. Two of these models are largely used and are simple to develop – the shrinking core model, and the progressive conversion model (LEVENSPIEL, 1974). They separate the leaching process in two large determining steps: the surface chemical reaction and the mass transport between the fluid and the solid. Considering these two models, this work aimed to study the leaching kinetics of roasted zinc concentrates in diluted solutions of sulphuric acid to determine the process-determining step and estimate the kinetic and thermodynamic parameters, such as specific chemical reaction rate, activation energy, apparent enthalpy, and entropy.

2 METHODOLOGY

2.1 SAMPLES

The samples used in the leaching experiments were produced by roasting zinc sulphide concentrates. Their chemical and crystallographic compositions are presented in Table 1 and Figure 1. The roasted concentrate was homogenized and duly quartered to obtain representative samples. After quartering, the sample was granulometrically separated (-65# +100#, -100# +150#, -150# +200#, and -200# +270#) by wet screening. The particle size distribution is shown in Table 2. Each granulometric fraction was washed with acetone for its quick drying.

Table 1 – Chemical composition of the roasted zinc concentrate.

Elements	% m/m	Elements	% m/m	Elements	% m/m
Zn	63	Pb	2.3	Mn	0.2
Fe	4.0	Ca	1.4	S	0.5
Cu	0.4	Cd	0.4	Others	27.8

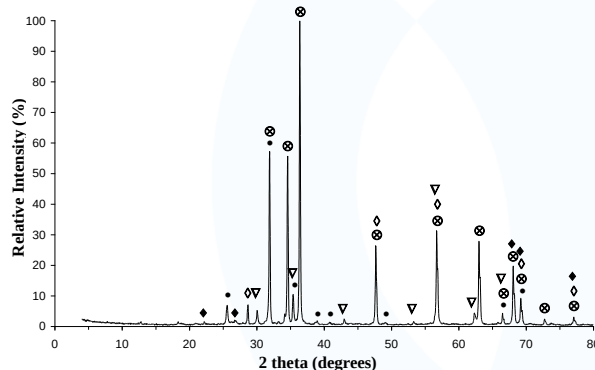
Source: Prepared by the authors

Table 2 – Particle size distribution of roasted zinc concentrate.^a %X is the percentage of the mass which passes through an aperture smaller than d^* .

Aperture size (d^* μm)	%X ($d < d^*$) ^a
210	54
149	44
74	14
43	5.6
37	5.0
< 37	4.5

Source: Balarini *et al.*, 2008

Figure 1 - XRD patterns of zinc concentrates (-400#) after roasting. ($\diamond$ sphalerite (ZnS); $\blacklozenge$ quartz (SiO_2); Δ anglesite (PbSO_4); # galena (PbS); $\circ$ pyrite (FeS_2); $\bullet$ willemite (Zn_2SiO_4); $\otimes$ zincite (ZnO); ∇ franklinite (ZnFe_2O_4)).



Source: Balarini *et al.*, 2008

2.2 LEACHING EXPERIMENTS

The discontinued leaching tests were performed in a 500-mL glass reactor with 6.5 cm i.d. and 15 cm high, an external jacket for the control of the temperature and a condenser couple to the lid to prevent water vapour loss. The stirring system was controlled with a mechanical agitator with digital control and a glass impeller with two pitched blades at 60° . The temperature range investigated was 30-70 $^\circ\text{C}$ and the sulphuric acid concentrations ranged from 4 g.L^{-1} to 10 g.L^{-1} . The mass of the roasted concentrate was calculated considering an acid consumption at the end of the experiments limited to 10% of its mass. At set intervals, reacted samples were taken, vacuum filtered, and diluted to prevent the chemical reaction continuing. The zinc extraction was determined by atomic absorption spectrophotometry in Thermo Electron Corporation, M Series. These results allowed calculating the percentage of zinc extraction as a function of time.

2.3 KINETIC MODELLING

The conversion data were modelled as a function of the time using kinetic equations of the shrinking core and progressive conversion models for the mass transport through the inert product layer (Eq. 1) and surface chemical reaction (Eq. 2) (Levenspiel, 1974).

$$\begin{cases} \frac{t}{\tau} = 1 - 3(1 - X_B)^{2/3} + 2(1 - X_B) \\ \tau = \frac{a \cdot \rho_B \cdot R^2}{6 \cdot b \cdot D_e \cdot C_A} \end{cases} \quad (1)$$

$$\begin{cases} \frac{t}{\tau} = 1 - \frac{r_c}{R} = 1 - (1 - X_B)^{1/3} \\ \tau = \frac{a \cdot \rho_B \cdot R}{b \cdot k_s \cdot C_A} \end{cases} \quad (2)$$

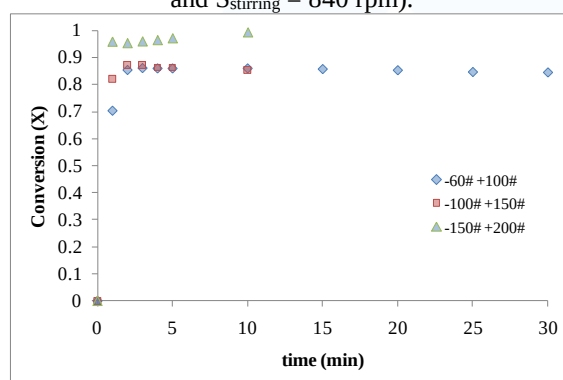
In Equations 1 and 2, t is the reaction time, τ is the time for the complete conversion of the particle, X_B is the conversion of the particle, D_e is the effective diffusion coefficient of the fluid reagent in the inert product layer, C_A is the acid concentration, a and b are the stoichiometry coefficients of the reaction ($aA_{(aq.)} + bB_{(solid)} \rightarrow \text{products}$), ρ_B is the molar density of reagent B in the solid, R is the particle radius, r_c is the unreacted core radius, k_s is the first order rate constant of the surface reaction.

3 RESULTS AND DISCUSSION

3.1 PRELIMINARY LEACHING EXPERIMENTS

To determine the operating levels of sulphuric acid concentrations for the present kinetic study, preliminary leaching experiments were performed for the different granulometric ranges given in Figure 2.

Figure 2 – Preliminary leaching experiments of the zinc roasted concentrate ($C_{\text{acid}} = 20 \text{ g.L}^{-1}$, $T = 30^{\circ}\text{C}$ and $S_{\text{stirring}} = 840 \text{ rpm}$).



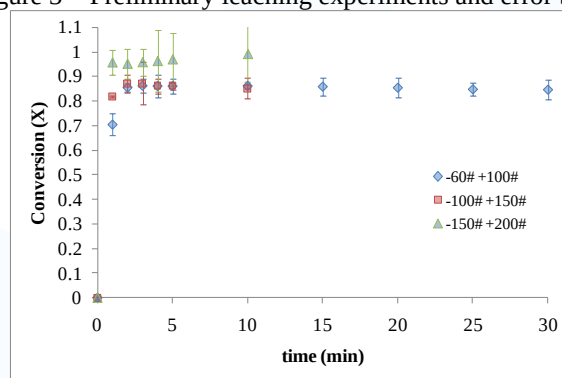
Source: Prepared by the authors.

Figure 2 shows that in the study conditions, the graph of conversion as a function of time levels off at times shorter than 5 min (1 or 2 min). Even at the lowest temperature level (30°C) and at the largest granulometry ($-60\# + 100\#$) investigated, the time for the complete conversion of the particles was around 2 min. This time shows that a concentration of 20 g.L^{-1} in the studied conditions would provide a high extraction rate, which would make obtaining the necessary experimental data for kinetic modelling impossible. It is very common in mineral and concentrate leaching experiments to obtain extraction curves with conversion lower than 100%, as shown in Figure 2, due to the presence of minerals that are more resistant to certain leaching conditions, such as temperatures lower than 60°C and diluted leaching agent solutions. In the case of roasted zinc sulphide, which was used in this work, Balarini *et. al.* (2008) observed the presence of ZnFe_2O_4 , zinc ferrite, whose structure is much similar to that of franklinite. This structure is very

stable and extremely resistant in mild leaching conditions. We point out that the determination of the zinc content in the samples requires acid attack for their complete acid digestion in extremely aggressive conditions (HCl and HNO₃ 3:1 (v/v) (*aqua regia*) at boiling temperature). In these conditions, all the zinc was solubilized, regardless of its structure. Therefore, the amount of zinc can be divided in a readily soluble portion and a leaching-resistant portion. From the three granulometries investigated (Figure 2), those with smaller particle diameter (-150# +200#) had the largest conversion rates.

Figure 3 shows the preliminary leaching results with the respective error bars. It shows that the experiment of the -150# +200# granulometry has the greatest experimental variability, and given its error range, it comprises the other two experiments, except for the time of one minute. Consequently, this graph does not allow stating that the extraction rate increased for the finest granulometric range. The -100# +150# range may already correspond to the optimal granulometry.

Figure 3 – Preliminary leaching experiments and error bars.



Source: Prepared by the authors.

The H₂SO₄ concentration levels were defined by estimation using the theoretical kinetic models. According to Equations 1 and 2, regardless of the controlling step, the time for the complete conversion of the particle (τ) is inversely proportional to the leaching agent concentration in the solution (C_A). Therefore, the estimation relation given by Equation 3 was used.

$$\tau = \underbrace{cte}_{constant} \times \frac{1}{C_A} \quad \therefore \quad \underbrace{\tau \times C_A}_{experimental} = \underbrace{\tau' \times C'_A}_{estimated}$$

$$C'_A = \frac{\tau \times C_A}{\tau'} \quad (3)$$

From Equation 3, one can estimate the concentration of the leaching solution (C'_A) by stipulating the time for the maximum conversion of the particle (τ').

The preliminary experiments showed that for a concentration of $C_A = 20 \text{ g.L}^{-1}$, the total conversion time of the particles was $\tau \approx 2 \text{ min}$ (Figure 2). Stipulating a total conversion time of $\tau' \approx 5 \text{ min}$, the estimated concentration would be $C'_A = 8 \text{ g.L}^{-1}$. Theoretically, with this time and sampling intervals of one minute five experimental points in the initial region of the leaching curve would be obtained. Longer times might be chosen; however, the acid concentration would be very low. Consequently, the sample mass of concentrate will be very small in order to keep the acid consumption nearly unaltered. As stated in section 2.2, the acid consumption was determined as being 10% m/m maximum. All models considered that the leaching agent concentration remained constant during the whole leaching experiment.

After determining the time for the maximum conversion of 5 min, a time of 10 min was used in all the following experiments, which theoretically is sufficient to understand the initial period and the plateau region. At this point, it should be highlighted that the time of 5 min needs to be validated.

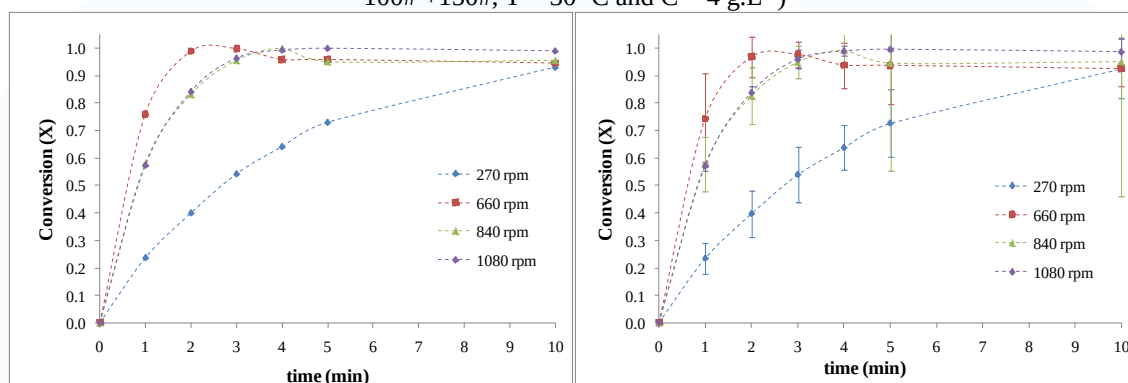
In the studies for the evaluation of the stirring speed, the temperature and the granulometric range on the Zn extraction, the acid concentration was 4 or 6 g.L^{-1} in order to expand the initial leaching period, which allow a better mathematical modelling of the kinetic equations.

3.2 THE EFFECT OF THE STIRRING SPEED

Many leaching processes, mainly those that use fine particles, require a level of stirring that allows both the suspension of the solid particles and the efficient mixture of the leaching solution (Dreisinger and Abed, 2002). Furthermore, it is possible to eliminate

the stirring speed as a variable in the kinetic study, as long as it is fast enough (Wadsworth, 1979). To this end, four leaching experiments at 270, 660, 840, and 1,040 rpm were performed. The results are given in Figure 4. In these experiments, the acid concentration used was 4 g.L^{-1} .

Figure 4 – Effect of the stirring speed (a) without error bars and (b) with error bars. (Granulometry = - 100# +150#, $T = 30^\circ\text{C}$ and $C = 4 \text{ g.L}^{-1}$)



Source: Prepared by the authors.

The analysis of Figure 4(a) shows that at high stirring speeds (660, 840, and 1,080 rpm), the extraction rate (initial curve slope) was greater, as expected. The similar behaviour for 840 and 1,080 rpm indicates that the boundary layer around the particles reached a minimum and that the stirring speed did not influence the extraction rate any longer. The same holds for 660 rpm. Apparently, the extraction rate in this condition is higher than for higher stirring speeds. However, the critical analysis of the experimental errors, Figure 4(b), shows that the error margin of the two first experimental points of the curve for 660 rpm comprises the experimental points of the curves for higher speeds, suggesting that these points may be equal. Likewise, the same happens for the last three points of the same curve. We suppose that in these conditions, from the stirring speed of 660 rpm, this parameter does not influence the leaching process any more. However, the limit that determines the influence of stirring speed on the zinc extraction may be smaller than 660 rpm, as no experiments were carried out in the 270 to 660 rpm range.

The paired samples t-test is another statistic method that could be used to determine if the leaching curves for 660, 840 and 1080 rpm are similar. For a fixed significance

level of 5% ($\alpha = 0,05$), i.e. a confidence interval of 95%, four comparisons were made (660 e 840 rpm, 660 e 1.080 rpm, 840 e 1.080 rpm e 270 e 1.080 rpm). The last comparison pair (270 e 1.080 rpm) was chosen to validate the information given by the paired samples t-tests, since both curves are visibly different (Fig. 3). The statistical results are presented in Table 3.

Table 3 – Paired samples t-tests for the kinetic leaching curves in different stirring speeds. Parameters for the statistical hypothesis tests ($\alpha = 0,05$ e $H_0: \mu_D = 0$).

Variable	660 rpm	840 rpm	Variable	660 rpm	1080 rpm
Mean	0.785	0.750	Mean	0.785	0.764
Variance	0.126	0.129	Variance	0.126	0.137
Observations	7	7	Observations	7	7
p-value (two tailed)	0.316		p-value (two tailed)	0.580	
Variable	840 rpm	1080 rpm	Variable	270 rpm	1080 rpm
Mean	0.750	0.764	Mean	0.495	0.764
Variance	0.129	0.137	Variance	0.097	0.137
Observations	7	7	Observations	7	7
p-value (two tailed)	0.119		p-value (two tailed)	0.006	

Source: Prepared by the authors

From the results of Table 3, all the three tests comparing the curves for 660, 840 and 1080 rpm present a p-value higher than 5%, which lead to a rejection of the null hypothesis ($H_0: \mu_D = 0$). This rejection indicates that the three curves (660, 840 and 1080 rpm) are statistically considered the same. As expected, the p-value for the pair 270 and 1080 rpm are lower than 5%, corroborating the difference between those two curves and validating the data of the paired samples t-tests.

According to the previous results, the stirring speed of 840 rpm were chosen as a fixed parameter for the subsequent experiments, since the effect of the mixing on the leaching rate, at this level, was surely eliminated.

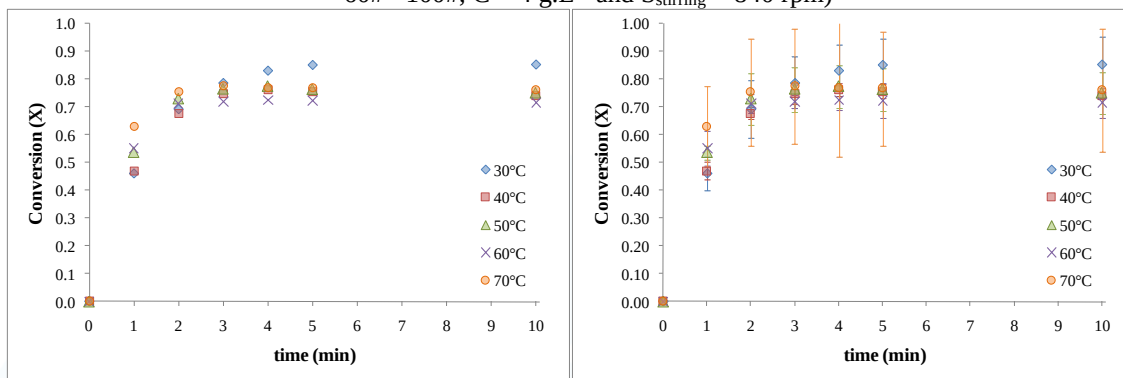
3.3 THE EFFECT OF TEMPERATURE

The dependence on the temperature may be used to estimate the activation energy and the apparent activation enthalpy and activation entropy (Dreisinger and Abed, 2002).

The activation energy is an extremely useful parameter for the identification of the process determining step; however, this is not consensual. According to Prosser (1996), values lower than 25 kJ.mol^{-1} characterize a mass transfer-controlled process (through fluid and/or product layer) and processes controlled by chemical reaction may have either lower or higher activation energy values. According to Levenspiel (1972) *apud* Dreisinger and Abed (2002), it is widely accepted that systems with activation energy higher than 40 kJ.mol^{-1} are controlled by chemical reaction (linear leaching), while those with activation energy less than 40 kJ.mol^{-1} are controlled by mass transport processes (parabolic leaching), whether in the product layer or a boundary fluid film.

Figure 5(a) shows that the temperature has a discrete effect on the leaching curves and that a small increase in the extraction rate (curve slope) can be noted at one min, when the temperature rises from 40 to 50°C and from 60 to 70°C. Furthermore, one can notice that all the curves, with the exception of that of 30°C, level off after 3 min of reaction. The slight difference between this curve and the others after 3 min of reaction and the small increase in the extraction observed at one minute are within the experimental errors, as shown in Figure 5(b). This figure also shows that the higher temperature curve (70 °C) also have comparatively larger experimental errors. This high data dispersion for 70 °C is due to the difficulty in controlling the system at this temperature. Despite the use of the condenser and jacket, the water vapour and heat losses increase with the increase in temperature.

Figure 5 – Effect of temperature on leaching (a) without error bars and (b) with error bars. (Granulometry = -60# +100#, $C = 4 \text{ g.L}^{-1}$ and $S_{\text{stirring}} = 840 \text{ rpm}$)

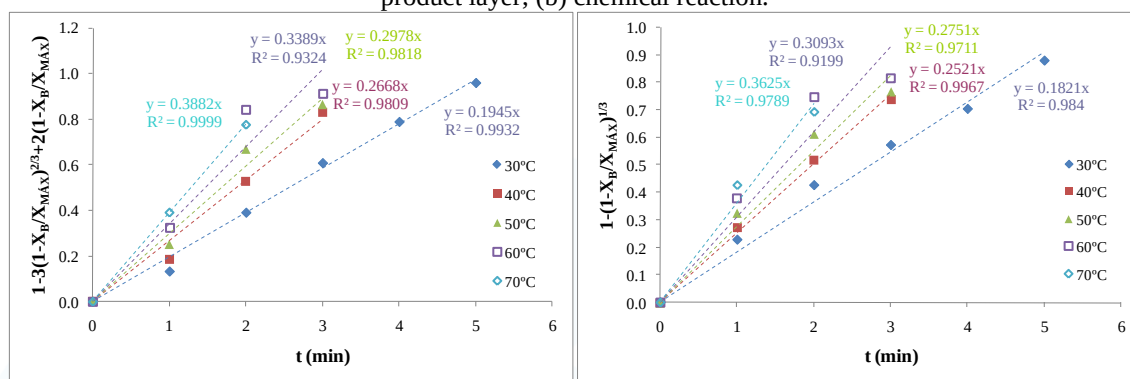


Source: Prepared by the authors.

The lack of significant differences in the reaction rate with the increase in the temperature shown in Figure 5 seems to indicate that mass transfer may be the process controlling step. One way to verify this hypothesis is to use theoretical kinetic models. Figures 5(a) and 5(b) display the linearization curves of the shrinking core model considering the diffusion in the product layer and the chemical reaction as the determining steps, respectively.

From the analysis of the determination coefficients (R^2) and of the curves in Figure 6, we can see that the data demonstrate a slightly better fitting for the control by the diffusion in the product layer than for the chemical reaction control. However, the presented linearizations are not sufficient to define the controlling step of the investigated system. The study of other variables, such as granulometry, acid concentration, which will be discussed below, are also required. Until other parameters are evaluated, both steps will be considered as controlling the leaching process.

Figure 6 – Linearization curves of the shrinking core model – effect of temperature: (a) diffusion on the product layer; (b) chemical reaction.



Source: Prepared by the authors.

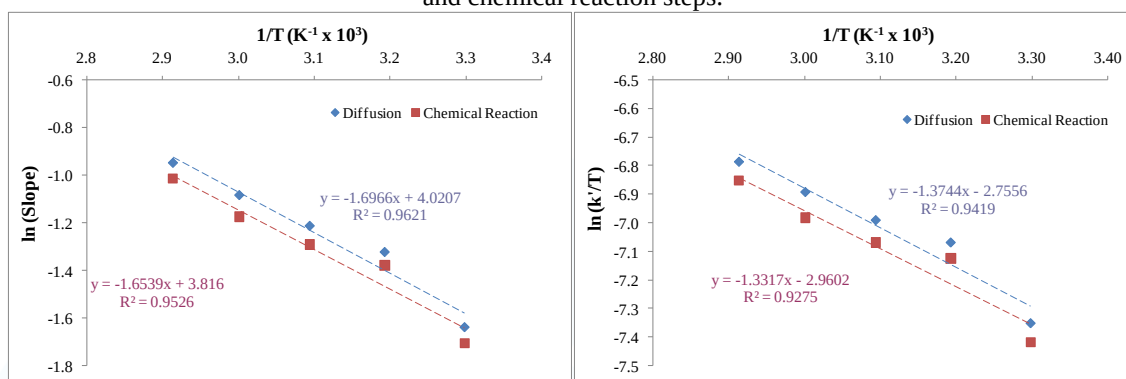
With the values of the slopes (**K**) of the curves in Figure 6, it is possible to determine the activation energy (**E_a**) of each controlling step, which is calculated with the Arrhenius Equation (Eq. 4) and determine the activation entropy and enthalpy using the equation of the theory of the transition state or the absolute reaction rate (Eq. 5) (Wadsworth (1979) *apud* Glasstone *et al.* (1941)).

$$\ln(K) = \ln(K_0) - \frac{E_a}{R} \times \left(\frac{1}{T}\right) \quad (4)$$

$$\ln\left(\frac{K}{T}\right) = -\frac{\Delta H^\ddagger}{R} \cdot \left(\frac{1}{T}\right) + \ln\left(\frac{k}{h} \cdot \exp\left[\frac{\Delta S^\ddagger}{R}\right]\right) \quad (5)$$

In Equations 4 and 5, parameter **K**, in the non-catalytic heterogeneous kinetics context, represents the slopes of all the linearized data, **K₀** is the pre-exponential factor or the factor of frequency, **E_a** is the activation energy in J.mol⁻¹ or cal.mol⁻¹, **R** is the universal constant of gases (8.14 J.mol⁻¹.K⁻¹ or 1.987 cal.mol⁻¹.K⁻¹), **T** is the absolute temperature in Kelvin, $\Delta H^\ddagger$ and $\Delta S^\ddagger$ are the apparent activation enthalpy and entropy, respectively, **k**, is Boltzmann's constant ($k = 1.38 \times 10^{-23}$ J.K⁻¹), and **h** is Planck's constant ($h = 6.63 \times 10^{-34}$ J.s). Figure 7 displays the Arrhenius and of the theory of transaction state curves.

Figure 7 – (a) Arrhenius curves; (b) Theory of transition state curves for the diffusion on the product layer and chemical reaction steps.



Source: Prepared by the authors.

From the data in Figure 7 were calculated the values of activation energy, frequency factor, activation enthalpy and entropy for the two controlling steps shown in Table 4. Both activation energy values given in Table 4 have the same order of magnitude and are below the values established as limits by Levenspiel (1972) and Prosser (1996) (40 kJ.mol^{-1} and 25 kJ.mol^{-1} , respectively). According to Prosser (1996), this results would be inconclusive to define the process controlling step, while by Levenspiel's (1974) analysis, the diffusion step would control the process (parabolic leaching), which agrees with the inference from Figure 5 earlier in this section. The work of Jandová *et al.* (1999) on the leaching of zinc compound in sulphuric acid solutions has results that are similar to the ones in this current study. These researchers studied several types of zinc oxide leaching in sulphuric acid solutions. The activation energy values calculated by Jandová *et al.* (1999) for the zinc oxide dissolution process (containing Al, Cr, Cu, Fe, Ni, Mg, and Mn) in 0.5 mol.L^{-1} sulphuric acid and temperature range of $40\text{-}60^\circ\text{C}$ were 7.9 kJ.mol^{-1} and 24.2 kJ.mol^{-1} . The difference between the values resulted from the calcination temperature used in the preparation of the oxides, 800 and 1000°C , respectively. Comparing the data in Table 4 with the results obtained by Jandová *et al.* (1999), the activation energy values of the diffusion and chemical reaction steps in this work were intermediate to those found by the latter. We point out they used synthetic zinc oxide and in the present work zinc oxide resulting from the roasting of a concentrate was used.

Table 4 – Thermodynamic parameters for different rate-controlling step.

	Diffusion	Chemical reaction
k_0	55.74	45.42
E_a (kJ.mol ⁻¹)	14.11	13.75
ΔH^* (kJ.mol ⁻¹)	11.43	11.07
ΔS^* (J.mol ⁻¹ .K ⁻¹)	-254.50	-256.20

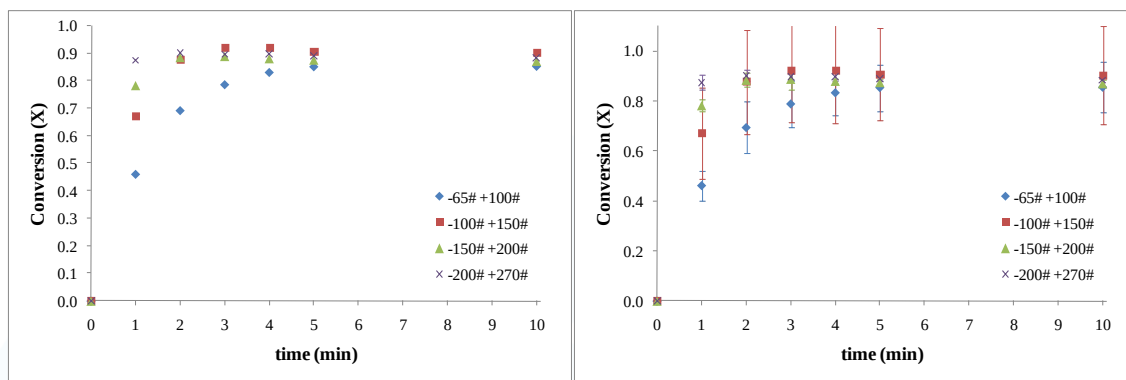
Source: Prepared by the authors.

From the data in Table 4, we can see for both controlling steps that the activation enthalpy values and the activation energy have the same order of magnitude. According to Wadsworth (1979), in hydrometallurgical systems, in which the temperatures are relatively low, the difference between the experimental activation energy (E_a) and the activation enthalpy (ΔH^*) is generally smaller than 1.0 kcal.mol⁻¹ (4.187 kJ.mol⁻¹). The values of E_a and ΔH^* also suggest that the leaching system must receive heat to facilitate the reactions (endothermic processes, $\Delta H^* > 0$). However, according to the behaviour of the extraction curves presented in Figure 5(a), the influence of the temperature on the system is relatively low, which explains the small order of magnitude of the activation energy ($E_a < 25$ kJ.mol⁻¹ (Prosser, 1996) and $E_a < 40$ kJ.mol⁻¹ (Levenspiel, 1972)). In relation to ΔS^* , the values found in the present work (-254.50 kJ.mol⁻¹ (diffusion model) and -256.20 kJ.mol⁻¹ (chemical reaction model) has the same order of magnitude of that found by Dreisinger and Abed (2002) (-212.4 J.mol⁻¹.K⁻¹).

3.4 EFFECT OF GRANULOMETRY

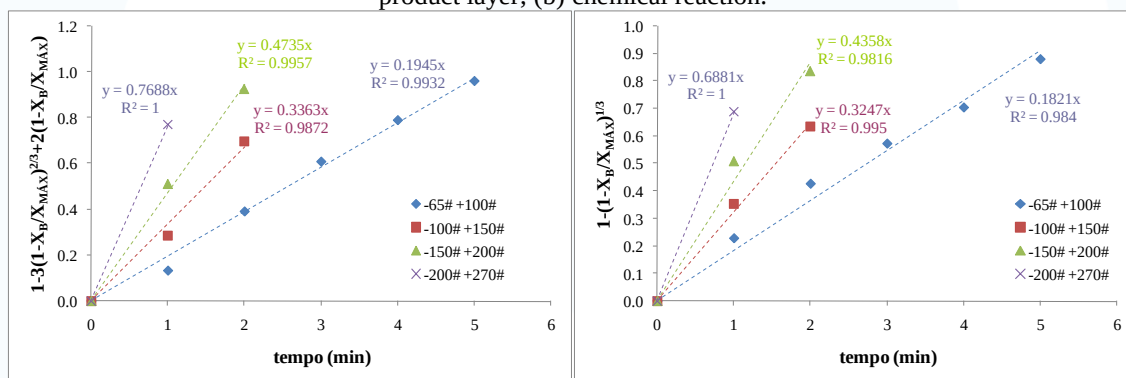
Another important parameter on leaching is the particle size (granulometry). The granulometry is directly related to the degree of liberation of the minerals present in the ores and concentrates. The curves of extraction as a function of the granulometric range are given in Figure 8. The analysis of the effect of this parameter also contributes to define the process determining step.

Figure 8 – Effect of granulometry on leaching (a) without error bars and (b) with error bars. ($T = 30\text{ }^{\circ}\text{C}$, $C = 6\text{ g.L}^{-1}$ and $S_{\text{stirring}} = 840\text{ rpm}$).



Source: Prepared by the authors.

Figure 9 – Linearization curves of the shrinking core model – effect of granulometry: (a) diffusion on the product layer; (b) chemical reaction.

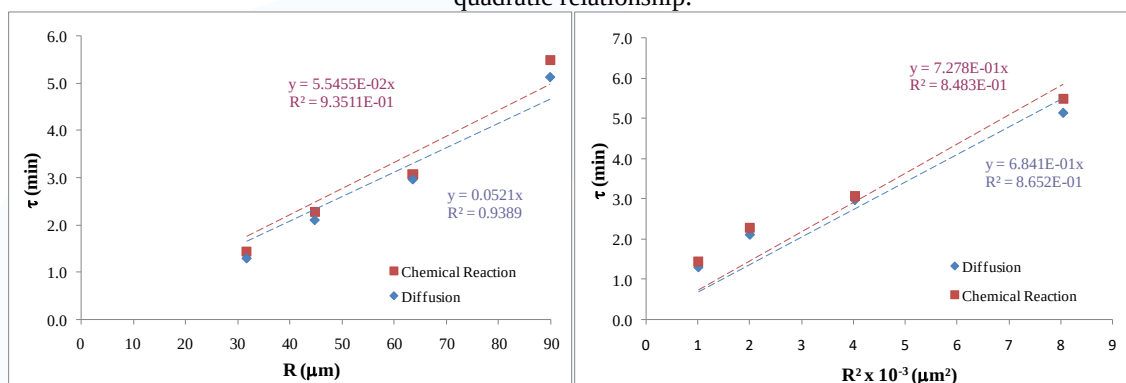


Source: Prepared by the authors.

Generally, the reduction in the particle size implies a considerable increase in the final conversion and an increase in the extraction rate due to the liberation of the mineral, as can be seen in Figure 8. Another important aspect that can be observed is that the decrease in the particle size did not result in an increase in the maximum conversion of the particle (around 90% for all curves), only a reduction in the time for the maximum conversion of the particle (τ). The value of τ decreased from 5 min for the larger particle size (-65# +100#) to 1.0 min for the smaller particle size (-200# +270#). According to Figure 8(b), we can see that the largest experimental errors were those of fraction -100# +150#. It is noteworthy that for this granulometry the experiments were carried out in

duplicate, while for the other fractions the experiments were done in triplicate. Figure 9 presents the modelling for the diffusion on the product layer and the chemical reaction.

Figure 10 – Correlation between the maximum conversion time and the radius: (a) linear relationship; (b) quadratic relationship.



Source: Prepared by the authors.

The analysis of the determination coefficients (R^2) alone did not allow to determine the controlling step, as in some granulometric ranges the linearization fits better (R^2 closer to the unity) considering the diffusion control, while in others the chemical reaction fits the results better. Another important point worth noting is the number of experimental points available for linearization. Due to the rapid leaching kinetics of the roasted concentrate, only the experiment of the thicker fraction (-65# +100#) allowed modelling more than three experimental points (Fig. 8). Consequently, the modelling of the other granulometric fractions was relatively compromised in respect to the linear behaviour.

As previously said, it is possible to identify the process controlling steps by the analysis of the value of τ in relation to the particle radius (R). This analysis was done for every curve and controlling step. The data are given in Figure 10.

Both linearizations, $\tau \times R$ and $\tau \times R^2$, must intercept the origin of the Cartesian plan, as the value of τ is zero when the particle radius is zero. It can be noted that although both the data given in Figure 10(a) and 10(b) can be represented by a straight line, when the origin interception condition is considered, a first order polynomial fits the experimental data given in Figure 10(a) better. This is an indication that the chemical reaction probably is the system determining step, as systems controlled by chemical reaction

present a linear behaviour ($\tau \propto R$), while systems controlled by diffusion on the product layer have a quadratic behaviour ($\tau \propto R^2$). We point out that this observation is not conclusive, as in the experiments with the thinner fractions the number of modelling experimental points was smaller than or equal to three (Fig. 8). Once the determination of τ depends on the slope of the linearized curves of the model, the errors resulted by this determination is much larger because of a lack of experimental data.

According to Jandová *et al.* (1999), the reaction products of the zinc oxide dissolution in sulphuric medium do not form an insoluble protective layer at the surface. Balarini *et al.* (2008) analysed the residues obtained from the digestion of the same roasted concentrate of the present work with a solution of sulphuric acid (50% v/v) and they observed the presence of tiny particles of quartz (SiO_2), sphalerite (ZnS), and anglesite (PbSO_4) but not the presence of the leaching product (ZnSO_4) what suggests that the product layer was not formed around the particles. As a result, the possible mechanism which physically describes the leaching of ZnO in sulphuric acid solution is the chemical reaction at the particle surface.

The constant rate for the first order chemical reaction (k_s) was estimated from the slope of the $\tau \times R$ curve given in Figure 10(a). According to this model, the relation between τ and R is given by Equation (6):

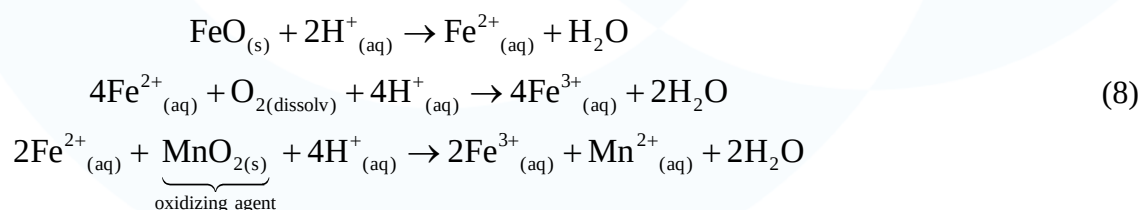
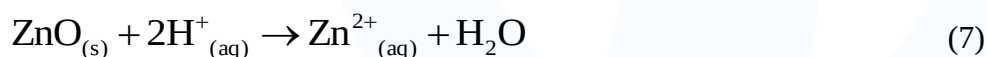
$$\tau = \underbrace{\left(\frac{a \cdot \rho_B}{b \cdot k_s \cdot C_A} \right)}_{\text{slope}} \times R \quad (6)$$

In this equation, **a** and **b** are the stoichiometry coefficients of the reaction of ZnO and H_2SO_4 (in this case, $a = b = 1$), ρ_B is the molar density of ZnO ($\rho_B = 6.92 \times 10^4 \text{ mol} \cdot \text{m}^{-3}$ (Perry and Green (1984))), and C_A is the concentration of hydrogen of H_2SO_4 (in this case, for $C_{\text{H}_2\text{SO}_4} = 6 \text{ g} \cdot \text{L}^{-1}$, $[\text{H}^+] = 6.89 \times 10^{-2} \text{ mol} \cdot \text{L}^{-1}$). With these data, **k_s** was estimated as approximately $3.00 \times 10^{-2} \text{ cm} \cdot \text{s}^{-1}$. According to the considerations of the theoretical models, the value of **k_s** is related to a first order reaction; therefore, it is necessary to

demonstrate if the modelling of the experimental data actually satisfies this condition. For such, the effect of the concentration of the acid on the extraction of zinc was studied.

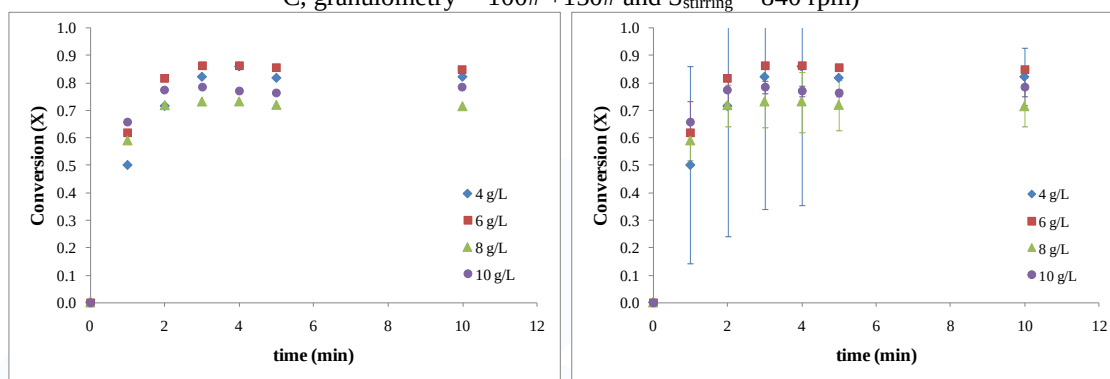
3.5 EFFECT OF THE ACID CONCENTRATION

The concentration of the leaching agent is directly linked to the chemical reaction rate. In the case of the leaching of roasted zinc concentrates in sulphuric medium, object of this study, the main chemical reaction can be represented by Equation (7). As the roasted concentrate has several other elements (impurities), parallel reactions of these components with the leaching agent may occur. An example is the ferrous oxide, which consumes H^+ and releases ferrous ions to the aqueous solution, and then Fe^{2+} reacts with dissolved oxygen from the medium and/or an oxidizing agent to raise its oxidation number from +2 to +3 using even more H^+ (Eq. (8)).



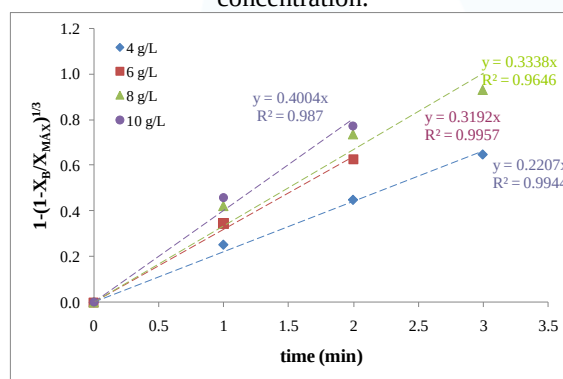
Other species than FeO can also consume H^+ ; however, the influence of these species will depend on their content in the concentrate. Figure 11 presents the curves of extraction for four concentrations of sulphuric acid (4, 6, 8, and $10 \text{ g}\cdot\text{L}^{-1}$).

Figure 11 – Effect of acid concentration on leaching (a) without error bars and (b) with error bars. ($T = 30^{\circ}\text{C}$, granulometry = $-100\# + 150\#$ and $S_{\text{stirring}} = 840 \text{ rpm}$)



Source: Prepared by the authors.

Figure 12 – Linearization curves of the shrinking core model (chemical reaction) – effect of the acid concentration.



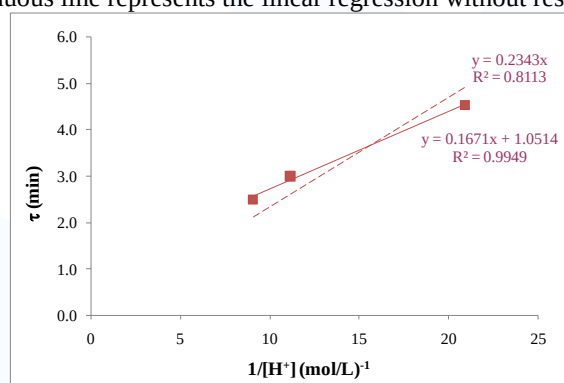
Source: Prepared by the authors.

Figure 11(a) shows a slight increase in the final conversion from the condition of 4 g.L^{-1} of H_2SO_4 to the condition of 6 g.L^{-1} , a decrease in the conversion from 6 g.L^{-1} to 8 g.L^{-1} , and an increase in the conversion from the latter to 10 g.L^{-1} . A more detailed analysis of the results (see Fig. 10(b)) shows that all the curves are within the confidence intervals with a significance level of 5% (except the curve for 6 g.L^{-1} , for which it was not obtained any replicate and which therefore does not have an error bars). One of the three replicates of the experiment of 4 g.L^{-1} of H_2SO_4 was lost, which explains why its error bar values were relatively high. Due to these problems, these results could not be conclusive. To obtain further information on the effect of the concentration of the acid on the extraction curves, the experimental data were modelled using the chemical reaction as the controlling step (Figure 12).

Differently from the initial observation made based on Figure 11(a), the analysis of the slopes of the models as a function of time shows an increase in the extraction rate of zinc with the increase in the concentration of the acid. It can also be observed that the slope of the curve of 6 g.L⁻¹ for the linearized data is similar to that of the curve of 8 g.L⁻¹. As this curve was plotted without replicate, it was eliminated from the following analyses.

According to Equation (6), the time for the maximum conversion of the particle (τ) is inversely proportional to the leaching agent concentration (C_A) ($\tau \propto 1/C_A$). C_A generally is the leaching agent species that is effective in the extraction of the solute (in this case H^+). $[H^+]$ is calculated from the total concentration of sulphuric acid using the acid and water ionization equations together with the mass and charge balances. The relationship between τ and $[H^+]$ are presented in Figure 13.

Figure 13 – Maximum conversion time of the particles as a function of the reciprocal of the hydrogen ion concentration. The dashed line represents the linear regression that crosses the origin, while the continuous line represents the linear regression without restriction.



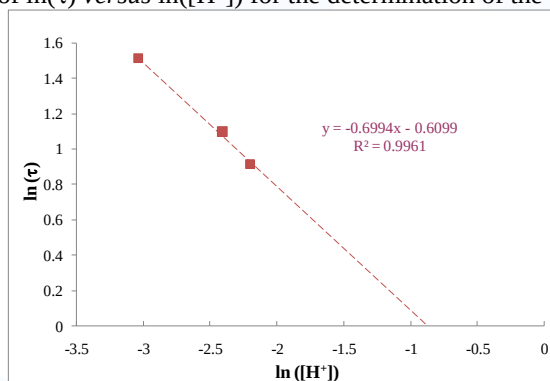
Source: Prepared by the authors.

In relation to the data in Figure 13, two types of linearization were performed, a linear regression crossing the origin (dashed line) and a free linear regression without restrictions (continuous line). Although in the first case the value of the determination coefficient $R^2 = 0.8113$ is relatively acceptable, the small number of experimental points for higher concentrations compromises a broader analysis. In the second regression, it was obtained a greater coefficient of determination than in the first case ($R^2 = 0.9949$); however, the presence of an additional term in the linear equation (constant **b** in the

polynomial $y = a \cdot X + b$) indicates it is not adequate for the physical meaning of the relationship between τ and $1/[H^+]$. Both regressions, with and without restriction, suggest a linear tendency for the correlation between τ and $1/[H^+]$, indicating that the chemical reaction between ZnO and H_2SO_4 can be considered a first order reaction in a first attempt. A way to directly evaluate the order of the reaction is the same used in Figure 14, based on the plot of $\ln(\tau) \times \ln([H^+])$. The slope of this curve corresponds to the exponent of the term $[H^+]$ (Figure 13).

From the value of the slope of the linear regression of Figure 14, the order of the chemical reaction was determined as approximately 0.7 (relatively close to the unity). The negative value means the inverse relationship between τ and $[H^+]$.

Figure 14 – Curve of $\ln(\tau)$ versus $\ln([H^+])$ for the determination of the order of the reaction.



Source: Prepared by the authors.

4 CONCLUSIONS

The present work evaluated kinetic aspects of leaching of roasted zinc in diluted solutions of sulphuric acid (4, 6, 8 and 10 g.L⁻¹). It was observed that the extraction of zinc is very rapid ($\tau \leq 5$ min), even in milder leaching conditions ($C_A = 4$ g.L⁻¹ and $T = 30$ °C). The experimental analysis using the shrinking core model and the progressive conversion model mathematically demonstrated that both diffusion on the ash layer and chemical reaction describe these leaching experiments. From the physical viewpoint, the particle size decreases with time, although small inert particles basically constituted of quartz (SiO₂), sphalerite (ZnS), and anglesite (PbSO₄) remained at the end of the

extraction. Thus, the chemical reaction described the leaching process in question more appropriately. It is noteworthy that the conclusion obtained from the kinetic leaching experiments was not restricted in the setting of mathematical equations, but also on a careful analysis of the phenomenon of solid-liquid extraction. The estimated thermodynamic parameter values were: $13.45 \text{ kJ.mol}^{-1}$, $11.07 \text{ kJ.mol}^{-1}$, and $-256.20 \text{ J.mol}^{-1}.\text{K}^{-1}$ for the apparent activation energy, and enthalpy and entropy, respectively. The small value of the activation energy suggested that the temperature had no substantial influence on the leaching of ZnO in sulphuric acid solutions. The analysis of the extraction curves as a function of the acid concentration showed that the overall chemical reaction has an order value of 0.7. With this result, the reaction can be considered as a first-order one and the estimated value of the chemical reaction rate constant (k_s) was $3.00 \times 10^{-2} \text{ cm.s}^{-1}$.

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